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Concept and preliminary structural analysis of a crater-covering dome for future lunar habitats

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The prospect of establishing a human presence on the Moon has transitioned from the realm of science fiction to an achievable goal. The long-term objective of the Artemis program is to establish a habitat on the Moon that would enable crews to remain on the lunar surface for extended periods. The developmental pathway for such facilities culminates in structures that are manufactured and constructed predominantly from materials sourced on the lunar surface, in alignment with the In-Situ Resource Utilization (ISRU) concept. This paper presents a conceptual lunar habitat that was created by covering 17 m diameter crater in the Mare Tranquillitatis with a structure made from a lunar regolith-based geopolymer. Five shapes of the covering lid were analysed, including: two concave domes with rises of 0.5 m and 1 m; a flat circular slab; and two convex domes with rises of 0.5 m and 1 m. Structural analysis was performed using the Finite Element Method, employing material data from existing literature as well as original strength tests of alkali-activated material based on lunar regolith simulants conducted by the authors. Each model of structure was subjected to dead loads and varying levels of internal air pressure. The numerical analysis revealed the advantages of concave-shaped structures, where internal pressure induced compressive stress within the cross-section, thereby mitigating the risks of air leakage and decompression of the habitat and taking advantage of material in which compressive strength is higher than tensile strength.

Keywords Extraterrestrial constructions, Habitats, In situ resource utilisation, FEM analysis, Geopolymer

The Apollo programme culminated in the first manned lunar landing on 20 July 1969, representing one of the most significant achievements in the history of humanity. However, only twelve astronauts have set foot on the lunar surface during a span of three years[1]. A hiatus in manned missions has persisted for a duration exceeding fifty years. Nonetheless, over ten agencies have continued to conduct or are currently engaged in unmanned missions on the Moon[2]. Among these, the most prominent is the ongoing Artemis program, which aims to facilitate the return of humans to our natural satellite and to subsequently establish a permanent base there[3]. This perspective directs research activities toward the economic implications that will arise from the exploitation of lunar resources, which will be viable with the establishment of adequate infrastructure. It is essential to consider the automation and autonomy of the extraction of natural resources from the lunar surface. Furthermore, the construction of structures capable of providing shelter for humans must be addressed. Such facilities should provide protection from external factors that pose a mortal threat to humans. The fundamental engineering challenge lies in creating a robust and secure partition that will effectively maintain atmospheric pressure within the facility, thereby enabling continuous human habitation without the need for specialised suits. The presented study introduces the concept of a covering lid for a lunar crater, designed to maintain atmospheric pressure conducive to human life within. The static load analysis includes variants of the adopted solution, taking into account realistic material parameters of the material produced in situ.

The main aspects of construction of a civil engineering structure on the moon

From a civil engineering perspective, the design process of lunar facilities is influenced by several critical factors that ultimately determine the final structural form. Key considerations include the intended use of the facility, which determines internal environmental requirements, its dimensions, the selection of building materials, and the construction technology employed, as well as the static, dynamic, and thermal loads that the structure will encounter.

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Categorisation of lunar habitats

Lunar structures can be divided, based on their function, into three main groups: (1) habitats, (2) storage facilities or shelters, and (3) supporting infrastructure, or, somewhat more generally, into two groups, namely habitats and other structures[4]. The role of habitats is to provide an environment that is suitable for human habitation, not only in terms of survivability under extreme conditions but also in terms of human needs for comfort, privacy, etc.[5]. In other structures, such a human-friendly environment is not necessary. Activities required to be carried out in these areas of the base can be performed by robots or astronauts equipped with a Extravehicular Mobility Unit (EMU).

So far, the only living space on the lunar surface has been the Lunar Module (LM) from the Apollo missions. However, this module cannot be classified as a habitat, as it was not designed to ensure a human-friendly environment but rather to enable short-term human survival. The LM had a volume of about 6.7 m^3 with the 'habitable' space for two astronauts having only approximately 4.5 m^3 and an interior pressure of approximately 330 hPa[6]. It is evident that conducting a multi-week mission under such austere conditions is not feasible.

In the literature[7], extraterrestrial structures are divided into three classes. Each subsequent class presents a higher level of technological advancement in human in-situ manufacturing capabilities.

Class I—pre-integrated structures, manufactured, assembled, and equipped entirely on Earth. They are fully functional and ready to use immediately. As such, they should be subjected to numerous tests prior to launch to identify and rectify any defects. However, Class I structures are constrained by limitations in size and weight due to the transport capacity of the launcher.

Class II—pre-fabricated structures consist of components manufactured entirely on Earth, with assembly occurring beyond Earth. The deployment and assembly of these prefabricated elements at the target location can be time-consuming and necessitate the use of robotic systems. Additionally, fitting the structurally prepared habitat requires a significant investment of time. Critical infrastructure, encompassing both equipment and structural elements, alongside the assembly procedure, must undergo a series of tests to identify and address any issues. Inflatable structures are of particular interest; they consist of a rigid sluice structure and an inflatable section made of flexible material that permits folding or rolling. Inflatable structures exhibit a favourable ratio of transport dimensions and weight to target volume[8], and their erection process is highly automated. However, the relatively thin walls of inflatable components may not provide sufficient protection against micrometeorites or cosmic radiation, and maintaining a stable internal temperature can be challenging. Consequently, an additional cover made of regolith is often proposed as an insulating and protective layer for this type of structure[9]. Class II structures face fewer constraints regarding weight and dimensions, as the dimensions of individual components can be appropriately adapted to the transport capabilities of the launcher. Moreover, the transportation of components can be segmented into multiple flights. An intriguing habitat structure was presented in[10]. Several inflatable tubes, arranged in either a longitudinal or circular configuration, are covered by regolith on their upper surfaces. Protection against radiation from other directions is afforded by additional embankment-like walls constructed from lunar soil, complemented by mirrors. This design allows for a substantial transparent area through which light may enter the habitat.

Despite the significant advancements in launch technologies within the space sector, the costs associated with such operations remain prohibitively high. Estimates suggest that, in the coming years, the expense of transporting cargo between Earth and the Moon will range between \$100,000 and \$1,000,000 per *kg*[11].

A natural consequence of this economic reality is Class III—structures primarily composed of materials sourced from the Moon according to concept of In-Situ Resource Utilization (ISRU)[12]. However, the production of materials suitable for construction may be a time-intensive process[13] and necessitates specialised manufacturing equipment (sieves[14], solar furnaces[15], lasers[16], etc.) and critical infrastructure that need to be transported from Earth. Domes and their derivatives are frequently proposed as the preferred geometric forms for this class[17–20]. Such a shape, when combined with a suitable selection of construction technology, enables the elimination of formwork or scaffolding and holds the potential for complete automation[21].

Another frequently suggested solution in the literature for future extraterrestrial constructions is the utilization or adaptation of natural landforms. Utilization of lava tubes, unique underground caves formed by volcanic eruptions[22], for habitation purposes is an example. This idea has been analysed for decades and contains two main concepts: unpressurised and pressurised lava tubes. In the first solution, the natural formation serves a role of shielding (protection against radiation and impacts of micrometeorites) and helps in thermal control but still needs an inflatable or modular structure inside for human habitation[23]. In the second one, ensuring the hermeticity of a system is the biggest challenge, but unquestionable advantages, such as large living space and leeway in functional area division, must be highlighted[24]. However, in both cases, potential locations for habitats are limited mostly to the formation of "sinuous rilles"[25], and because of the depth of lava tubes (in[26] the depth of the investigated lava tube was estimated to be between 80 and 88 m) the building process and transportation are potentially more challenging than operating on the lunar surface. The next idea, adopting lunar craters or rilles for habitation purposes, is some kind of compromise between surface and lava tube or cave solutions. The usage of these formations is advantageous in terms of reducing construction material and simpler groundworks[27], when compared to surface structures of similar size and similar regolith layers for protection. An example of the concept of locating an egg-shaped facility in a natural depression is presented in[28]. Moreover, because of the numerous craters of various morphology, it is plausible to find an optimal one for future habitation in scientifically or commercially significant locations.

On the basis of the described pros and cons of the various structure types, the authors endorse the use of Class III.

Building materials

The utilization of lunar regolith, the most abundantly available material on the Moon, as a primary component in construction materials is highly justifiable. The regolith exhibits variable depths; in[29], researchers estimated the average thickness to be approximately 10 m in the mare and 12 m in the highlands. However, these values may be significantly variable. For example, in the analysis of approximately 44.000 km^2 of China's Chang'E-5' landing area, the thickness of regolith ranged from 0.74 to 18 m with a mean of 7.15 m[30]. Lunar missions have enabled extensive studies of regolith samples obtained from both mare and highland regions, demonstrating that their chemical and mineralogical compositions are comparable to those of terrestrial basalts and anorthosites, respectively. The chemical compositions of samples collected during various missions can be found, for example, in[31]. Table 1 presents a summary of the average oxides content found within the lunar regolith.

Due to the limited quantity of lunar regolith samples retrieved and returned to Earth, predominantly acquired during the Apollo missions, which amount to approximately 380 kg[32], research into potential building materials is conducted using regolith simulants (analogues). An exemplary list of simulants and their manufacturers can be found in the resource provided by the National Aeronautics and Space Administration (NASA)[33]. However, this list notably omits several significant simulants, including the CAS-1 lunar soil simulant[34], the ALRS-1 Australian Lunar Regolith Simulant[35], DNA-1 De NoArtri[36], the NAO-1 National Astronomical Observatories simulant[37], and the Lunar Regolith Analogue AKG-2010[38]. The various analogues differ in terms of chemical composition, mineral composition, and physical properties[39], and they do not fully replicate the characteristics of authentic lunar regolith.

Material research to date has encompassed a range of innovative materials, including geopolymers[40,41], polymer concretes[42], sulfur concretes[43], and cement concretes, for which the finest fractions of regolith can serve as Supplementary Cementitious Materials (SCMs)[44]. Additionally, there have been proposals to develop building materials incorporating substances derived from biological sources such as blood[45] and urine[46], as well as approaches utilizing bacteria for binding[47]. The literature also discusses methodologies that rely on the stabilization of regolith without the use of supplementary binders, such as sintering[48] or melting[49]. Although several studies on materials have been conducted in vacuum chambers to simulate extremely low-pressure conditions[50], or in climatic chambers where suitably low[51] or high temperatures are maintained[52], these experimental conditions do not fully replicate the environment present on the lunar surface, primarily due to differing gravitational interactions. Similarly, the necessity of utilizing more or less reliable lunar regolith simulants in place of actual lunar regolith leads to the conclusion that findings derived from research on lunar building materials must be approached with considerable caution.

Construction technology

One of the proposed technological solutions for the construction of Class III structures is 3D printing, which may involve the sintering of layers of regolith through powder bed fusion (PBF) or the extrusion of material derived from lunar regolith. In PBF technology, the structure is incrementally created layer by layer by using thermal energy, typically from a laser source, to irradiate and fuse particles within a powder bed[53]. In extrusion-based 3D printing, components are fabricated through the sequential layering of a paste material, which subsequently hardens[54,55]. There are already operational prototypes of this type of printer[21,56]. Certain proposed 3D-printed structures can be constructed without the necessity of additional support[57], while the incorporation of, for example, inflatable formworks results in aesthetically appealing shapes and forms of structure[58].

The construction solution adopted, the erection technology, and the conditions on the Moon themselves determine the properties that the building material should posses, both during the construction phase, the maturation phase, and when fully mature. In 3D printing, the consistency of the extruded paste must exhibit sufficient ease of pumping, uniformity of structure, adequate cohesion to adjacent layers, and the ability to retain shape under its own weight and that of subsequent layers[56,59]. Furthermore, these properties must be maintained for the required duration[60]. Moreover, low shrinkage, creep, and thermal expansion are expected during the setting stage and for the fully mature material, as well as sufficiently high strength. High thermal insulation, low gas permeability (mainly to oxygen), and the ability to absorb cosmic radiation are also desirable properties.

Another crucial aspect of construction technology is the foundation method employed to prevent potential undesired displacement of structures. In the case of certain constructions, such as protective walls and inflatable structures, it is sufficient to remove sharp stones and perform site levelling before placing them directly on the lunar surface[61]. For other, more rigorous structures such as habitats, the solutions for foundations must

All in (%)	Mare	Highland
SiO_2	45.4	45.5
Ti_2O_2	3.9	0.6
Al_2O_3	14.9	24.0
FeO	14.1	5.9
MgO	9.2	7.5
CaO	11.8	15.9
Na_2O	0.6	0.6

Table 1. Average percentage oxide content of lunar regolith samples[31]. Fe_2O_3 included in FeO total.

be more advanced. In[62], the authors identified anchoring as a fundamental foundation method for modular extraterrestrial structures to prevent uplift and avoid the necessity for deep excavation.

Extraterrestrial conditions and requirements for habitats structure

The gravity on the Moon is approximately $\frac{1}{6}$ g, and the atmospheric/exospheric layer is exceptionally thin[63], permitting the assumption that the Moon can be regarded as essentially a vacuum. The absence of an atmosphere, which on Earth is a reservoir of oxygen and functions as a thermal stabiliser as well as a natural barrier against micrometeorites and radiation, contributes to the extreme environmental conditions observed on the Moon.

In the context of a structure designed for long-term human habitation, the aspect of comfort will be required. This comfort can only be assured if the internal pressure of the facility approximates that of Earth. Otherwise, occupants would need to wear protective suits. Given the evident oxygen deficit and the necessity for pressure generation in large-volume facilities, it is imperative to consider maintaining a reduced pressure that will not induce health issues (both physical and mental) in the long term[64,65].

For this reason, the facility's tightness will be a priority issue and is another element that, beyond the obvious mechanical strength, will determine the safety level of the entire engineering structure. It is noteworthy to highlight a particularly interesting approach in which the internal action of the pressure is balanced by the dead loads from a regolith layer of appropriate thickness, which functions as ballast for the structure. Assuming a complete absence of pressure on the Moon and disregarding the mass of the structural and covering layers of the facility, the thickness of the regolith required to achieve equilibrium at an internal pressure of 1000 hPa would range from 28 to 42 m, contingent upon the bulk density of the soil[66], which can range from 1550 to 2250 kg/m3. When maintaining a pressure of 600 hPa, which corresponds to an air pressure at an elevation of 4000 m above sea level on Earth, the requisite regolith thickness would be reduced to a range of 17 to 25 m. It is also essential to consider transient pressure states resulting from construction phases, accidental decompression[67], planned pressure reductions[68], and other influencing factors.

One significant environmental threat to the structures on the lunar surface is posed by micrometeoroids, which are meteorites measuring up to a few centimetres in size. Due to the absence of an atmosphere, the Moon is subjected to a significant frequency of micrometeoroids impacts. The average amount of dust and micrometeorites is estimated to be approximately 1.4 t/day[69]. As found by[70], about 80% of meteoroids contributing to this flux is smaller than 10 cm. Additionally, as also pointed out by[71], the velocities of the background population of lunar impactors range from 5 to 65 km/s. These extreme velocities and consequently highly energetic impacts can lead to the melting or even vaporisation of regolith[72]. Such energetic impacts have the potential to cause local damage to civil structures or even immediate catastrophic failure.

Another issue of lunar facilities in respect to the terrestrial ones is their need of shielding against cosmic radiation. Structures on Earth designed for long-term human occupancy do not require extra protection because the presence of an atmosphere and a magnetic field serves as a built-in shield[73]. There are two main types of man-dangerous cosmic radiation, high-energy electromagnetic waves (gamma rays and X rays) and high-energy particles (protons and nuclei). Particles that come to our solar system from interstellar space are referred to as Galactic Cosmic Rays (GCR). GCR consists of nuclei of essentially all elements of the periodic table, but mainly hydrogen and helium (comprising more than 98 %)[74] with energies reaching up to 10 GeV and average fluxes of approximately 4 particles $\mathrm{cm}^{-2}\mathrm{s}^{-1}$[75]. Radiation from the inside of our solar system is emitted by the Sun. The solar wind (SW) consists of ions where hydrogen and helium nuclei are the most abundant (approx. 99.9 %)[76]. The SW flux is approximately 10^8 particles $cm^{-}\ s^{-1}$, with particle velocities typically ranging from 300 to 600 km/s and energies generally between 0.5 and 2 keV[77]. At this time, it is worth paying attention to the modulation of GCR by SW. The more intense the SW, the less radiation from out of our solar system (GCR) enters to our heliosphere. These fluctuations of SW intensity are observable in 11- and 22-year solar cycles[78]. The last but not least radiation phenomena that is dangerous for astronauts is Solar Particle Events (SPEs). SPEs are relatively short (from hours to days) but very energetic and intense emission of mainly protons by the Sun[79]. Particles originating from SPEs are supersonic, with energies ranging from 10^1 to 10^6 keV[80]. All the aforementioned particles that originate from space are called primary particles. These primary particles have the ability to interact with matter present in the lunar soil, shelter, or the human body. As a result of these interactions, secondary particles including protons, alpha particles, beta particles gamma rays, X-rays, neutrons, and heavy charged particles are produced[81]. Both primary and secondary particles are ionising radiation that can, at the microscopic level, lead to complex and difficult to repair DNA damage responsible for degenerative processes, cell death, or long-term cancer mutations[82]. During the most severe SPEs, the radiation dose absorbed by the human body within a matter of hours may result in Acute Radiation Syndrome (ARS), which may be fatal[83]. Space agencies such as NASA, the European Space Agency (ESA), the Canadian Space Agency (CSA), the Japan Aerospace Exploration Agency (JAXA), and the State Corporation for Space Activities "Roscosmos" establish their own criteria concerning occupational limits for the career effective dose and the short-term (30 days) dose for astronauts. The most significant information regarding dose and risk assessment for crews from various space agencies operating on the International Space Station (ISS) is presented in[84]. According to the aforementioned study, the career effective dose limit for the whole body varies among agencies, ranging from 0.5 Sv (JAXA for females aged 27 at first exposure) to 0.6 Sv (NASA), and extending to 1.0 Sv (ESA, CSA, "Roscosmos," and JAXA for males older than 45 years). For short-term exposure, organ-specific equivalent dose limits have been established. For instance, all partners of the ISS have reached a consensus on a limit of 25 mSv for blood-forming organs. This concludes that radiation protection plays a crucial role in protecting the health of astronauts during long-term missions and there are several ideas of protection. The first is nutritional[85] or pharmacological supplementation[86,87]. The second is active radiation shielding subdivided into plasma, electrostatic, and the most developed magnetic shield[88]. The third one is passive shielding of habitats in which looking for optimal material seems to be crucial[89,90]. Aforementioned ideas might be combined to obtain hybrid radiation protection[91] and habitats interior might be divided into several zones with different protection levels. In[92] the researchers proposed the establishment of a dedicated area, specifically a shelter within the habitat, designed to provide significantly enhanced protection against radiation. This shelter would serve as a refuge for astronauts during periods of intense radiation events.

Proposal of the Moon's crater cover structure

The authors analysed the concept of a covering lid for a lunar crater, designed to maintain an atmospheric pressure conducive to human life inside, made from ISRU lunar regolith based material.

Construction site

The criteria for selecting a site for a lunar habitat must consider a range of factors. In[93] proximity to the placement of solar panels, proximity to scientifically significant locations, proximity to recoverable materials for ISRU, and the availability of a certain amount of flat terrain for landing pads have been identified as valuable considerations. Furthermore, the authors emphasise the potential advantages of natural topographical features in terms of thermal control and radiation protection, as well as their role in reducing the quantity of materials required and minimising excavation efforts.

Studies of over 2700 small craters located in four different areas selected to be representative of the entire Moon have demonstrated that, in general, highland terrains exhibit a lower crater density than mare regions[94]. However, this density remains significantly high, with hundreds of craters/km^2. The referenced paper confirms the existence of craters with a diameter of approximately 20 m and a depth exceeding 8 *m*. This fact prompted the authors of this paper to employ a hypothetical small crater for the analysis of habitat structure. The approximate dimensions of the crater are 17 m in diameter and 6 m in depth (Fig. 3f), and it is located in the Mare Tranquillitatis, near the Apollo 11 landing site (latitude 0.67° North and longitude 23.47° East in the selenographic coordinate system—Fig. 1).

From the civil engineering perspective, geotechnical parameters of crater soil and crater morphology, along with the temperature profile, are among the most important factors that highly influence the structural solution of the habitat. However, in this paper, the analysis of these factors was reduced to general assumptions of the required diameter and depth of the crater.

Building material

In light of previously mentioned experience and test results, it is feasible to propose the implementation of a structure made from lunar regolith based geopolymer (GP). The term GP is most frequently used to describe synthetic, inorganic polymers that are primarily composed of aluminium and silicon. Although geopolymers have been known and used to a limited extent in construction since the 1950s, there has been a notable surge in interest over the past decade. This can be attributed to their distinctive mechanical properties, which are analogous to those of cement concrete, and the scientific community's pursuit of a material that can replace Portland cement as the foundation for the most prevalent construction material, concrete[96–98]. As indicated in Table 1, the lunar regolith contains an average of 45 % SiO_2 by weight. This finding suggests that the material includes pozzolanic compounds, which are predominantly associated with the amorphous phase content (influence of thermal history), soluble silica content, and particle size[99]. It is noteworthy that the inclusion of a significant proportion of filler is advantageous, as it decreases the amount of material that must be transported from Earth to the lunar surface.

The geopolymer samples presented in this study consisted of three components: a 10 molar sodium hydroxide (NaOH) solution, prepared by dissolving solid sodium in distilled water; sodium silicate water glass (Na_2O x $nSiO_2$ x nH_2O), which collectively functions as the activator; and LHS-1[100], produced by Exolith Lab in the USA, the most widely accepted lunar highland regolith simulant, which serves as the filler.

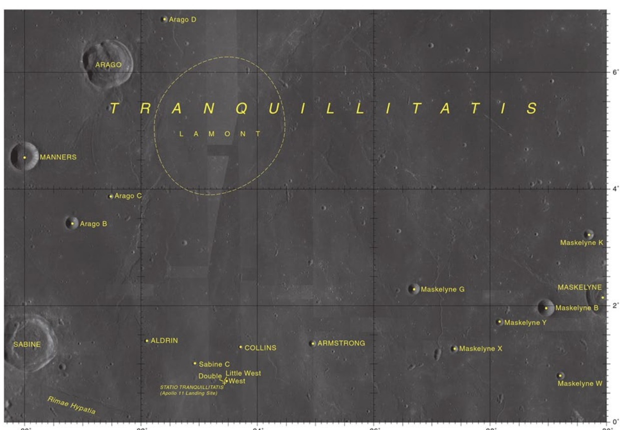

Fig. 1. Location of the site for the analysed structure—a hypothetical crater situated near a latitude 0.67° North and a longitude 23.47° East within the selenographic coordinate system[95].

An activator consisting of a mixture of sodium hydroxide and water glass in a ratio of 1:2 was combined with LHS-1 and mixed for a duration of five minutes. Assuming a weight of NaOH equal to one part, the samples were formulated using ten parts of LHS-1. The resulting paste was subsequently cast into plastic triple moulds measuring $40 \times 40 \times 160$ *mm* (see Fig. 2a).

The curing conditions to which the samples were subjected were selected to simulate lunar conditions. The surface temperature of the Moon, in the analysed location, reaches values of nearly 120°C and can drop to approximately -180°C. However, within the span of 7 terrestrial days, the temperature does not fall below $60^{\circ}C$[101,102]. This temperature range is conducive to the geopolymerisation process. According to the results of tests published in e.g.[103,104] the temperature above 60°C accelerates geopolymerisation and increases the initial strength. Assuming that the preparation of the paste and the initial stage of curing could take place during the above-mentioned 7-day period, the temperature of $60^{\circ}C$ used for the tests guarantees a safe approach. Based on the aforementioned considerations, the samples were subjected to curing in a thermal vacuum chamber at a temperature of 60°C and a pressure of 50 hPa (Fig.2). The 50 hPa pressure represented the minimum achievable pressure within the apparatus, effectively simulating the vacuum conditions present on the Moon. Following a curing period of one week, the specimens were demoulded, and the samples continued to cure under consistent conditions for the next 3 weeks.

After a total curing period of 28 days, bending and compressive strengths were evaluated in accordance with the standards outlined in EN 196-1:2016. A three-point bending test was conducted to ascertain the tensile strength in bending, after which the compressive strength was assessed. Prior to conducting the strength tests, the beams were measured and weighed to establish their volumetric density. It is important to note that the samples placed in the vacuum chamber are subjected to a decompression process. The reduction in pressure resulted in an increase in volume for the geopolymer samples under analysis. Consequently, the average volumetric density of the material that underwent hardening under conditions of reduced pressure was determined to be 1750 kg/m^2. Tensile strength of samples was $f_t = 5.5$ MPa while the compressive strength was $f_c = 9.0$ MPa. The f_t/f_c ratio was approximately 60 %. A comparable compressive strength value is reported in[105], where alkali-activated regolith simulant samples were evaluated under similar conditions. In the referenced study, the samples were also subjected to curing under reduced pressure; however, the curing temperature was slightly elevated, reaching 90°C. In addition to assessing material strengths, scanning electron microscopy (SEM) and X-ray diffraction (XRD) analysis was conducted. The X-ray diffraction patterns suggest labradorite as the crystal binding phase in the geopolymer (Fig. 2b). The Scanning Electron Microscopy image of the geopolymer structure was presented in Fig. 2c.

One of the significant properties of a material is its elasticity, which is characterised by Young's modulus. In accordance with the Eq. (1) proposed by[106], the modulus of elasticity for geopolymers may be estimated based on the results of compressive strength tests.

$$E = 0.037\rho^{1.5}\sqrt{f_{ck}} \tag{1}$$

where E is the Young's modulus (*MPa*), f_{ck} is the compressive strength (*MPa*), ρ is the unit weight (kg/m^3).

The Young's modulus of the analysed lunar regolith simulant-based geopolymer, as derived from Eq. (1), is E=8.13 GPa. This value is significantly lower than that of structural concrete, which exhibits Young's modulus of 27 GPa for C12/15, 36 GPa for C45/55, and 44 GPa for C90/105, as stated in EN 1992-1-1. However, it is substantially higher than that of autoclaved aerated concrete masonry, which possesses a modulus of elasticity ranging from approximately 1.5 to 5 GPa[107–110]. In relation to the most comparable material concerning this parameter, the authors identified masonry cement-sand calcium-silicate, which exhibit a modulus of elasticity of approximately 7.8 to 8.5 GPa[111,112].

It is noteworthy that lunar regolith is regarded as a viable material for radiation protection shielding. In[113], the authors estimated the dose inside a lunar shelter composed of 10 cm of aluminum and different thicknesses of lunar regolith during a solar event similar to the very intense February 1956 one. The estimated relative

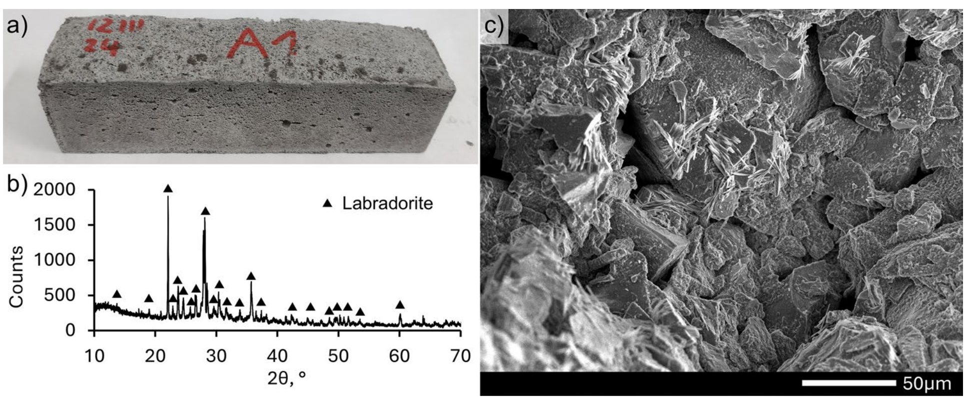

Fig. 2. The geopolymer sample: (**a**) 40 × 40 × 160 mm specimen; (**b**) X-ray diffraction pattern (XRD); (**c**) scanning electron microscopy image (SEM).

attenuation was 15 % for a 50 g/cm^2 regolith shield, 43 % for a 70 g/cm^2 regolith shield, and 70 % for a 140 g/cm^2 shield. The level of 50 mSv (recommended for radiation workers) required the use of a 145 g/cm^2 regolith shield in the analysed case. The authors pointed out that for the same mass thickness, lunar regolith is slightly more effective than aluminum for shielding energetic solar protons. The 10 *cm* layer of aluminum may be replaced by a regolith layer of 27 g/cm^2. Similarly, the simulation of shielding against radiation of "lunamer" (regolith-based geopolymer) presented in[114] indicated that a 50 cm thick layer is sufficient for mean radiation during a prolonged (12-month) lunar mission. However, the authors emphasize that, for rare solar flares containing high-energy protons, other methods of protection against radiation should be provided.

As the permeability of the proposed geopolymer has not been tested yet, the authors suggest the usage of an inflatable structure inside that serves as an "inner tube" for habitat hermeticity.

Geometry and loads

In the presented study, the authors aim to identify the optimal solution for the crater cover (Fig.3g) from the perspective of structural mechanics. Five different configurations of the covering structure (A-E) for a 17 m crater were analysed.

- A concave dome with a radius of R = 48 m, resulting in a negative rise of 1.0 m (Fig.3a).
- B concave dome with a radius of R = 94 m, resulting in a negative rise of 0.5 m (Fig.3b).
- C flat plate in the shape of a circle (Fig. 3c).
- D convex dome with a radius of R = 94 m, resulting in a positive rise of 0.5 m (Fig.3d).
- E convex dome with a radius of R = 48 m, resulting in a positive rise of 1.0 m (Fig.3e).

The thickness of each analysed structure was assumed to be 0.5 *m*. Assuming a volumetric density of 2050 kg/m^2 for the material, it revealed that the lowest demand is associated with case C (flat slab), which requires 49.64 *t* for the construction of this type of structure. The incorporation of a dome as a cover in the B or D configuration (rise of 0.5 m) resulted in a material consumption of 49.79 *t*, representing a marginal increase of 0.3 % compared to the slab configuration. The material consumption for cases A and E (rise of 1.0 *m*) was estimated to be approximately 50.23 *t*, reflecting an increase of 1.2 % relative to case C.

The authors focused on the static analysis of the covering structure; however, aspects such as tightness, radiation protection, and resistance to micrometeorite impacts were not addressed. Furthermore, it was posited that the solution utilised for establishing the foundation of the analysed structure would facilitate a seamless transfer of load to the ground. To standardise the analysed cases, it was assumed that the structure possesses no perforations, and solution of the entrance is not the focus of this paper. The principal objective is to perform an analysis of the covering structure in two distinct states:

- during the construction phase, where gravitational forces are the sole consideration—Condition "0",
- during the operational phase, where both gravitational forces and internal pressure, ranging from 100 to 1000 hPa, are taken into account—Condition 100 to 1000.

Description of the finite element method model utilised in the analysis

A numerical analysis of five crater cover shapes was conducted using the Finite Element Method (FEM) in Abaqus software.[115]. The authors used 4-node, quadrilateral, stress/displacement shell finite elements (FEs)

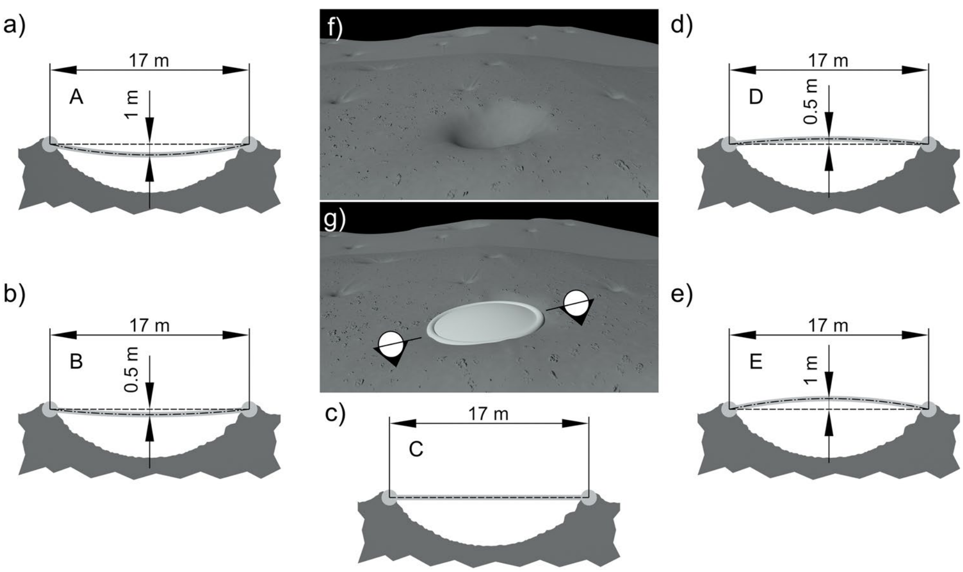

Fig. 3. Structure of crater cover: (**a**–**e**) cross section of a covering structure—case A-E; (**f**) crater; (**g**) visualisation of the crater's cover structure.

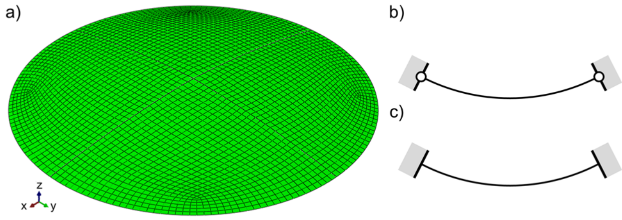

Fig. 4. Computational models employed in the analysis: (**a**) 3D shell FEM model—-sample A; (**b**) boundary condition with unrestricted rotation—**HR**; (**c**) boundary condition with unable rotation - **FR**.

Condition [hPa]	Model A		Model B		Model C		Model D		Model E	
	HR	FR	HR	FR	HR	FR	HR	FR	HR	FR
0	−0.03	−0.04	−0.16	−0.12	−2.76	−0.30	−0.16	−0.12	−0.03	−0.04
100	0.20	0.22	0.99	0.72	16.71	1.80	0.99	0.72	0.20	0.22
300	0.65	0.73	3.31	2.39	55.66	5.99	3.31	2.39	0.65	0.73
600	1.34	1.51	6.78	4.91	114.09	12.27	6.78	4.91	1.34	1.51
1000	2.25	2.53	11.41	8.26	191.98	20.65	11.41	8.26	2.25	2.53

Table 2. Values of vertical displacements of the central node (cm) from models A–E with different boundary condition—Fig.4.

denoted as S4R in Abaqus, each possessing six degrees of freedom at the nodes of the spatial structure model. The geometry for the designated cases A-E was a spherical cap with varying rises, as detailed in "Geometry and loads". Furthermore, it was assumed that the group of FEs would maintain a constant size of 0.3 m throughout the entire model area (shown in Fig. 4a). These assumptions are intended to enable a more thorough analysis of the structure's critical stress-strain state and the subsequent optimisation. The interaction between the foundation and the subsoil was not analysed in detail, with the authors presuming a permanent connection between the cover structure and the lunar soil. However, in future investigations, this issue must not be overlooked; the challenges posed by the variability of the crater crown's level and local discontinuities at the edge necessitate a comprehensive analysis. In the analysis two approaches were employed to consider the boundary conditions resulting from the structure being supported by a ring foundation. The first approach (the **FR** model—Fig.4c) involved a fixed connection between the structure and the foundation (no translational or rotational displacement at the edge of the structure); the second approach (the **HR** model—Fig.4b) involved a pinned connection between the structure and the foundation (only rotational displacement could occur at the edge of the structure).

The area of the modelled structure was subdivided into FEs, primarily in a quadrilateral configuration, with an average side length of 0.3 m. The objective was to achieve a uniform distribution of FEs, thereby mitigating the potential for stress concentrations that could arise solely from its geometric shape. The numerical model comprised between 2924 and 3104 FEs (Fig. 4a), which corresponded to a total of 3128 to 3312 nodes, each possessing six degrees of freedom. This configuration required a resolution of up to 20,000 equilibrium equations during the static analysis.

The assumption of linear elastic behaviour of the material was adopted from "Building material" and considered a conservative approach. It was therefore posited that the occurrence of non-elastic deformation may lead to material damage, potentially resulting in the loss of structural integrity. Furthermore, calculations were conducted on the basis of characteristic values, which means that material properties and loads are not subject to any safety factors.

Results of the numerical analysis and discussion

As a result of the conducted analysis, the values of vertical displacements measured at the node located at the central point of the structural model were obtained. This observation is applicable to all conditions involving internal pressures ranging from 0 to 1000 hPa, as well as the methodology employed in defining boundary conditions. The results are presented in Table 2, which consider certain cases due to linear variability, thus indicating a proportional relationship between internal pressure and vertical displacement at the central point.

It is important to note that the influence of gravity on the analysed structure of the crater roof is minimal. Model C (ver. **HR**) exhibited a maximum vertical displacement of -2.76 cm (the negative value indicating a downward movement), which corresponds to only 0.016 % of the span. The remaining cases yielded values that were 10 to 100 times smaller.

The increase in interior pressure resulted in the observed "lift" of the structure. This indicates that the effect of gravity was dominated by the pressure load even at 100 hPa. When the pressure was elevated to 300 hPa (equivalent to the pressure at an altitude of Mount Everest), Model C (ver.**HR**) exhibited a vertical displacement (upwards) of 55.66 cm (3.3 % of the span). Failure to account for the potential of support rotation in this model (ver.**FR**) resulted in a displacement value that was ten times smaller, measuring 5.99 cm, which constitutes 0.35 % of the span. A further increase in pressure to 600 hPa (equivalent to the altitude of Mont Blanc) led to a proportional increase in vertical displacement in Model C (ver.**HR**) to 6.7 % of the span (over 114.1 cm) or 12.27 cm (0.72 % of span) for Model C with ver. **FR**. Ultimately, simulating internal pressure at the level of normal atmospheric pressure for a human resulted in a vertical displacement in Model C of nearly 2 m (11.3 % of the span) for ver. **HR**. Models A and E (both ver. **HR** and **FR**) demonstrated displacements that, in absolute terms, corresponded to the displacement caused by the gravitational load of Model C ver. **HR**. The results presented underscore the importance of considering the method of connecting the structure with the crater crown in order to mitigate any delamination resulting from the effects of internal pressure. This fact cannot be overlooked by engineers tasked with designing structures on the lunar surface.

The most critical aspect from a safety perspective is the utilization factor of structure during the use of the object. Consequently, the extreme values of the principal stresses in the internal and external layers of the analysed numerical models necessitate further discussion. The limiting values of the properties of the material were adopted, corresponding to those obtained in the laboratory tests presented in " Building material", which are, respectively, the tensile strength $f_t = 5.5$ MPa and the compressive strength $f_c = 9.0$ MPa.

As with the displacements, the numerical models were examined in terms of the maximum tensile stress (Table 3), which represents a quantitative analysis contingent upon the specific cases studied (taking into account versions **HR** and **FR**). In scenarios where only lunar gravity was considered (condition "0"), no significant stress values were observed in the models. Model C exhibited the highest tensile stress, reaching a value of up to 1.2 MPa for ver. **HR**, while the remaining models displayed values that were between 3 to 100 times smaller. This indicates a minimum safety margin of 78 % based solely on tensile stresses, in instances where only the dead weight was taken into account, without the introduction of internal pressure within the crater. Considering the application of a pressure of 100 hPa, it is observed that the tensile stress in model C (ver. **HR**) exceeds the permissible limit, reaching a value of 7.24 MPa. This indicates that cracks are likely to form, which can compromise the airtightness of the building partition. Consequently, the flat cover is deemed unsuitable as an option for effectively sustaining a higher internal air pressure. It is noteworthy that the remaining models still exhibit a substantial safety margin, with an approximate value of 80 % (except model C ver. **FR** - 51 %). Model A (ver. **HR**) warrants particular attention, as in the 100 hPa model case, it exhibits a reduction in tensile stress by 50 % compared to the 0 *hPa* model case. This decrease results from opposing directions of gravitational forces and the simulated internal pressure. Increasing the pressure within the crater to 300 *hPa* did not surpass the tensile stresses observed in the models featuring a curvilinear structure. This indicates that for objects designed to maintain partial atmospheric pressure, crater covers constructed as concave or convex may be properly. However, it is important to note that the stress values for the structural models D and E reached approximately 70 % and 40 % (for ver. **HR**) of their respective tensile strengths, which presents a potential risk of leakage from the partition under such stress conditions. The effort achieved in the case where rotation of the supports is not possible (**FR** version) reaches 53 % in Model E and is borderline in Model D. This is corroborated by the results obtained at elevated internal pressures, which induce shell stress in models D and E that exceeds the material's assumed limit value. This indicates that the design of the crater covering, featuring a concave basin, is optimal if the internal pressure is anticipated to be comparable to that of the Earth. It is noteworthy that in model A, the stress was at a level of 11 % (ver. **HR**) in relation to the tensile strength. The influence of boundary conditions on the magnitude of tensile stresses and the resultant strength limit in Model A of the FR version is indeed evident.

The comprehensive stress analysis also included tests of numerical models with respect to maximum compressive stress (see Table4) in the quantitative assessment of the cases analysed. A result of 0.0 MPa indicates that the entire analysed area of the shell is in a state of tension. In cases where only lunar gravity was considered (designated as condition "0"), no significant stress values were observed in the models. Models D and E exhibited compressive stress, reaching values below 0.0, while the remaining models demonstrated values of 0 (indicating a tension cross-section). When a pressure of 100 hPa was applied, compressive stress was noted in models A and B (both ver. **HR** and **FR**), and was at least 10 times smaller than the threshold for the considered material. The solutions derived from models C, D, and E in the FR version indicate the presence of compressive stresses, which were not detected in the HR version. It is crucial to emphasise that elevating the air pressure to terrestrial levels did not result in the generation of compressive stresses in Models C to E, which were subjected to free rotational boundary conditions. This observation suggests that cracks are likely to form, which may compromise the integrity of the building envelope. Any cracks that develop within the structure could potentially widen, thereby threatening the overall tightness of the building. In contrast, such a phenomenon is not observed in Models A and B, where compressive stresses are present and, in the extreme case in Model B (ver. **HR**), exceed

Condition (hPa)	Model A		Model B		Model C		Model D		Model E	
	HR	FR	HR	FR	HR	FR	HR	FR	HR	FR
0	0.10	0.15	0.19	0.27	1.20	0.47	0.04	0.18	0.01	0.08
100	0.05	0.48	0.26	1.13	**7.24**	2.83	1.15	1.64	0.61	0.88
300	0.18	1.59	0.86	3.74	**24.10**	**9.42**	3.82	5.46	2.04	2.92
600	0.37	3.25	1.76	**7.67**	**49.40**	**19.30**	**7.84**	**11.20**	4.18	**5.98**
1000	0.62	5.47	2.96	**12.90**	**83.10**	**32.50**	**13.20**	**18.80**	**7.03**	**10.10**

Table 3. Values of the maximum tensile stress within the structure (MPa); bold is value higher than $f_t = 5.5$ MPa.

Condition (hPa)	Model A		Model B		Model C		Model D		Model E	
	HR	FR	HR	FR	HR	FR	HR	FR	HR	FR
0	0.00	−0.01	0.00	−0.03	0.00	−0.08	−0.15	−0.10	−0.07	−0.06
100	−0.40	−0.37	−0.91	−0.63	0.00	−0.48	0.00	−0.19	0.00	−0.08
300	−1.33	−1.24	−3.02	−2.10	0.00	−1.60	0.00	−0.64	0.00	−0.27
600	−2.74	−2.55	−6.18	−4.30	0.00	−3.28	0.00	−1.30	0.00	−0.56
1000	−4.60	−4.29	**−10.00**	−7.24	0.00	−5.52	0.00	−2.19	0.00	−0.94

Table 4. Values of the maximum compressive stress within the structure (MPa); bold is value lower than $f_c = -9.0$MPa.

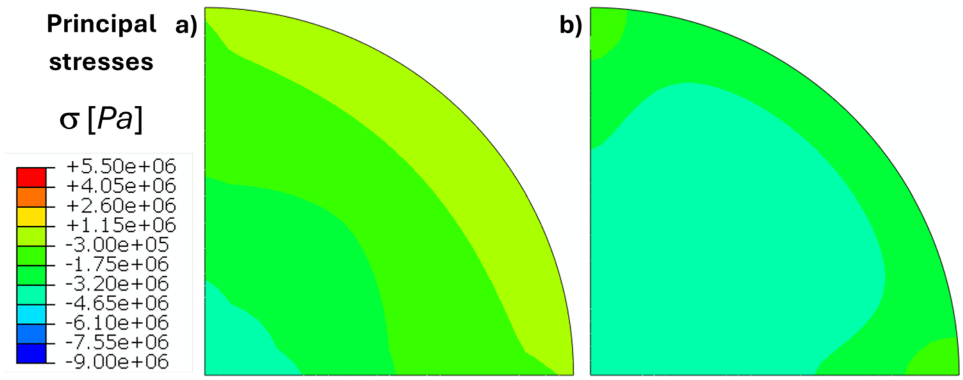

Fig. 5. Map of principal stress from a quarter of Model A (ver. **HR**) at 1000 hPa: (**a**) internal side; (**b**) external side.

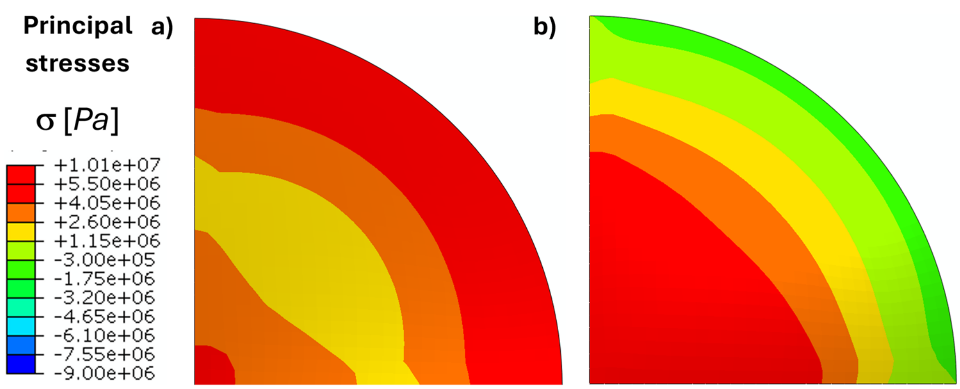

Fig. 6. Map of principal stress from a quarter of Model E (ver. **FR**) at 1000 hPa: (**a**) internal side; (**b**) external side.

the critical compression threshold. Simulating an internal pressure of 1000 hPa in Model A results in a stress level of 51 %. Consequently, the quantitative assessment of the stress state of the crater cover structure supports the implementation of a concave shape with a rise of 1.0 m. Relation $\frac{1}{17}$(concave arrow / cover diameter), which, when considering the presented results, ensures the requisite load-bearing capacity of the structure. It is important to note that in models D and E, an increase in the value of the convex arrow (by 0.5 m) results in a reduction of compressive stresses by 50 %. This trend is concerning, as it may lead to the potential occurrence of a cross-section of the structure experiencing full tension, which could compromise its overall integrity.

In Figs. 5 and 6, a quantitative analysis is presented, offering a comprehensive illustration of the stress state within the crater cover structure. Considered the distribution of stress on both the inner and outer surfaces, accounting for the limiting values of the material under tensile (red) and compressive (blue) conditions. An examination of the solution for Model A (ver. **HR**) under an internal pressure of 1000 hPa yielded the distribution of principal stresses, as depicted in Fig.5, for the inner (a) and outer (b) surfaces, limited to one quarter of the structure. It should be noted that none of the regions exhibit values that surpass the material's strength. Furthermore, the distribution of tensile stresses is observed within the support area on the inner surface, whereas it is absent on the outer surface. Taking into account the impact of compressive stress distribution and the quantitative results presented in the Table 4, it is noteworthy that altering the method of modelling the support rotation does not significantly influence the solution. Future analyses of the structure should consider modifications involving an increased cross-section and potential alterations to the curvature in the support area. This will lead to a reduction in the tensile stress values observed in Model A with the FR variant (see Table 3). However, such geometric alterations may result in the generation of compressive stresses on the outer surface of the shell, which serves as an inherent seal for this partition, thereby safeguarding the interior of the object against possible decompression.

The hazardous stress state of the crater cover structure, resulting from the qualitative analysis, is evident in the solution of Model E (ver. **FR**) under conditions of internal pressure at a level of 1000 hPa. The distribution of the principal stresses, as illustrated in Fig. 6, was obtained for a quarter section, revealing that in the central region, throughout the thickness of the cross section, the tensile strength of the material has been exceeded (indicated in red). This finding suggests that potential damage in this area may occur due to tearing and sudden decompression of the interior of the structure, posing a direct threat to the safety of the individuals within. A possible mitigation strategy for such a stress state could involve increasing the dead weight or the layer of backfill above the crater cover. Given the low gravitational forces and the convex geometry of the structure, it would be necessary to cover it with a substantial layer, estimated to be approximately 15 m thick. However, it must also be acknowledged that this layer might slide toward the crown, thereby exposing the vulnerable section. In this regard, the structural design of the crater cover raises concerns about its suitability for practical application.

Numerical analysis facilitates the evaluation of the potential critical stress-strain state on the lunar surface under varying internal pressure conditions. It is essential that the models developed accurately reflect the geometry and materials intended for the construction of future protective coverings. This aspect necessitates thorough consideration. Concurrently, investigations into the effects of micrometeorite impacts and cosmic radiation are warranted, representing additional avenues for the authors' ongoing research. At this stage, it is noteworthy that the incorporation of a concave curvature in a structure designed to withstand elevated internal pressures presents a viable solution that may be subject to optimisation in future studies.

Conclusions

The resurgence of crewed lunar missions, the establishment of a permanent base on the lunar surface, and the subsequent commercial utilisation of lunar resources pose significant challenges for engineers in terms of construction using the In-Situ Resource Utilisation concept. In the present study, lunar regolith simulant-based geopolymer samples were fabricated and cured under reduced pressure conditions. The compressive and bending tests conducted on the specimens support the consideration of real lunar regolith geopolymer as a viable structural material. The proposed composition of the geopolymer has a relatively high regolith filler to alkali activator ratio, which is desirable from a cost reduction perspective.

A structural analysis of a conceptual lunar habitat has been performed. In the proposed solution, a 17 m diameter crater was covered by a lid made from the aforementioned geopolymer. Five cases of the crater covering structure were examined: a concave dome with negative rises of 1.0 m and 0.5 m; a flat circular slab; and a convex dome with positive rises of 0.5 m and 1.0 m. Computational models of cases were subjected to dead loads and varying levels of internal air pressure. The findings of this analysis demonstrate the superiority of concave domes relative to the other geometric configurations when subjected to internal air pressure loads. The cross-section of the concave-shaped structure, characterised by a negative rise of 1.0 m and subjected to an internal pressure of 1000 hPa, exhibited a compression stress state on both the internal and external surfaces that remained below the material's compressive strength.

The proposed solution of a concave-shaped dome presents several advantages. Utilising natural landforms as part of the structure reduces the volume of material that needs to be produced. The compression stress state in the cross-section of the concave dome is favourable due to the mechanical characteristics of the material in question, which exhibit higher compressive strength than tensile strength. Moreover, the absence of decompression in the cross-section reduces the risks of gas permeability. The potential locations suitable for this kind of structure are significantly more numerous than for structures utilising lava tubes, which is desirable in the context of future utilisation of lunar resources. Operating in craters is also easier than operating in deep lava tubes. The incorporation of regolith as a component of construction materials provides a significant degree of protection against radiation and micrometeorite impacts, factors that must be carefully considered in the planning of long-term human settlements on the Moon, thereby favouring the authors' solution over, for example, inflatable structures. Furthermore, the inclusion of an additional layer of loose regolith above the structure is beneficial in terms of the previously described pressure equilibrium and is intriguing from a civil engineering perspective. However, the proposed solution for the concave dome habitat structure also has disadvantages, such as the reduction in inner height caused by the negative rise.

The authors acknowledge that the studies presented in this paper, encompassing both material tests and numerical analysis, are in a preliminary stage. Nonetheless, the simplified approach employed herein has proven adequate for conducting a preliminary comparison of various structural shapes and for identifying the most promising candidate for subsequent, more sophisticated and accurate analyses and tests. Future materials research must concentrate on several key aspects, including protection against radiation, permeability, thermal and thermo-mechanical properties, fatigue, and rheological behaviour. Additionally, hybrid testing under combined thermal-mechanical loads and high-velocity impact experiments should be conducted. Similarly, the simplifications within the computational model, such as the assumption of perfectly stiff support and a linear material model, should be noted. The necessity for more comprehensive analysis, particularly concerning a more accurate material model and its parameters, foundation solutions, soil-structure interaction, construction technologies and stages, as well as micrometeorite impacts, thermal flow, or vibrations, is unquestionable. The authors' ongoing research is focused on some of the aforementioned topics.

Data availability
The datasets utilised and/or analysed during the current study are available from the corresponding author upon reasonable request.

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Declarations

Competing interests
The authors declare no competing interests.

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